

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks available |     |     |       |       |      |
|----------|-----|------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 3        | (a) | (i)  |  | A – Oxygen – Diffusion/ simple diffusion<br>B – Glucose - Facilitated diffusion/ active transport<br><br>1 mark for correct substances<br>1 mark for transport mechanisms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1               | 1   |     | 2     |       | 2    |
|          |     | (ii) |  | <ul style="list-style-type: none"> <li>• {Oxygen/ (substance) A} passes through {phospholipid/ bilayer} because it is non-polar (1)</li> <li>• {Line A/ oxygen/ diffusion}: {linear relationship/ positive correlation/ directly proportional} (between concentration gradient and rate of diffusion)/ or description (1)</li> <li>• {Glucose/ (substance) B} uses {channel/ carrier/ intrinsic/ transmembrane} proteins as it is polar (1)</li> <li>• {Line B/ glucose/ facilitated diffusion/ active transport}: the rate is eventually limited by the number of {channel/ carrier/ intrinsic/ cell membrane} proteins (1)</li> </ul> Must refer to carrier proteins if use active transport | 2               | 2   |     | 4     |       |      |
|          | (b) | (i)  |  | <b>Mark points in pairs</b> <ul style="list-style-type: none"> <li>• No {DNA/ genes} (1)</li> <li>• to code for haemoglobin (1)</li> </ul> OR <ul style="list-style-type: none"> <li>• No ribosomes (1)</li> <li>• for translation/ protein synthesis. (1)</li> </ul> OR <ul style="list-style-type: none"> <li>• No Golgi {body/ apparatus} (1)</li> <li>• so no post-translational modification (1)</li> </ul> OR <ul style="list-style-type: none"> <li>• No mitochondria (1)</li> <li>• to make ATP for protein synthesis (1)</li> </ul>                                                                                                                                                   |                 |     | 2   | 2     |       |      |